2311018

” ” Word Embedding

1. Bag-of-Words Word2Vec/GloVe BERT
2.
3. Logistic Regression
4. Word2Vec GloVe
5. BERT Fine-tuning
6. Accuracy Macro-F1 Macro-F1
7.

2.1

/		
	Windows 10/11 (x64)	
CPU	Intel Core i5, 4 8	CPU
	16 GB	
Python	3.12.7 (Anaconda)	
PyTorch	2.13.0+cpu	CUDA: False
Transformers	5.x	BERT
scikit-learn	1.9.0	
gensim	4.3.x	Word2Vec
NLTK	3.9.x	

2.2

NYT - `text` - `label` sports/politics/business

AG News text Word2Vec

2.3

- 80% 9,215 - 20% 2,304 - 42 - stratified split

3.1

NYT - 42 - 20% -

3.2

1.
2. `\b\w+\b` tokenize
3. sklearn train_test_split

4.1 Accuracy

4.2 Macro-F1

Macro-F1 F1 Score F1 Macro-F1

Macro-F1 Accuracy

sports 75% Accuracy business politics sports Accuracy 75% Macro-F1

Task 1 Bag-of-Words 30

5.1

Bag-of-Words " "

5.2

5.2.1 Binary Bag of Words Vocabulary |V| |V| - x = 1 - x = 0

5.2.2 Word Frequency Binary BoW Word Frequency - x = c c i

5.2.3 BoW **Logistic Regression** - max_iter = 1000 - random_state = 42

5.3

	Accuracy	Macro-F1
Binary Bag of Words	0.9848	0.9634
Word Frequency	**0.9870**	**0.9687**
Count Vector	0.9861	0.9677

61,021

Word Frequency BoW 61,021

5.4

1. NYT 75% BoW ”score” “game” ”government” “election”
2. BoW 61,021
3.

Task 2 Word Embedding 50

6.1

Word Embedding 100-dimensional word vectors 100

n e(w), e(w), ..., e(w)

e(d) = (1/n) × Σ e(w)

6.2 Pre-trained GloVe glove.6B.100d

GloVe Embedding glove.6B 100-dimensional GloVe word vectors

- http://nlp.stanford.edu/data/glove.6B.zip
- 400,000
- 100

6.3 Word2Vec Trained on AG News

AG News 100-dimensional Word2Vec Model - Gensim Word2Vec - 100 - 5 - 1 - 10 - 90,000 AG News

6.4 Word2Vec Trained on NYT

NYT 100-dimensional Word2Vec Model - 6.3 - 9,215 NYT

6.5

	Accuracy	Macro-F1
GloVe (Pre-trained)	**0.9813**	**0.9573**
Word2Vec (AG News)	0.9774	0.9492
Word2Vec (NYT)	0.9783	0.9484

6.6

1. **GloVe** 40
2. NYT Word2Vec AG News
3. ” ” —— GloVe AG News
4.

Task 3 BERT 20

7.1

Fine-tuning - google-bert/bert-base-uncased - Tokenization max_length = 64 - 3 epochs

7.2

1. **padding** batch batch
2. **tokenize** BERT
3. epoch Macro-F1 checkpoint

7.3

Epoch	Train Loss	Val Macro-F1
1	0.2786	0.9403
2	0.0457	0.9403
3	0.0152	0.9417

7.4

	Accuracy	Macro-F1
BERT-base-uncased max_len=64, 3 epochs	0.9766	0.9469

7.5

1. **BERT** BERT Accuracy 0.9766 / Macro-F1 0.9469 Task 1 0.9870 / 0.9687
2. **max_length=64** BERT 49 8.0% 604
3. + 9,215 BERT-base 1.1 0.2786 0.0152 2 epoch
4. **BERT** " " BERT 49 Macro-F1 0.9469 160

7.6

"BERT " + Logistic Regression K

K	Test Accuracy	Test Macro-F1	
10	0.8993	0.8147	
20	0.9314	0.8781	
40	0.9523	0.9044	
49	0.9557	0.9114	BERT
80	0.9696	0.9367	
160	0.9748	0.9491	
320	0.9800	0.9594	
604	0.9896	0.9771	Task 1
BERT	0.9783	0.9509	49

1. 2. K = 49 BERT Macro-F1 0.9114 BERT 0.9509 4 3. BERT

8.1

			Accuracy	Macro-F1
Task 1	Binary BoW	57,453	0.9848	0.9634
Task 1	Word Frequency	57,453	**0.9870**	**0.9687**

			Accuracy	Macro-F1
Task 2	GloVe (Pre-trained)	100	0.9813	0.9573
Task 2	Word2Vec (AG News)	100	0.9774	0.9492
Task 2	Word2Vec (NYT)	100	0.9783	0.9484
Task 3	BERT (Fine-tuned)	768	0.9766	0.9469

8.2

1. Word Frequency Macro-F1 0.9687 BERT 0.9469 2 3
2. **Macro-F1 Accuracy** BERT 5 Accuracy 0.98 Macro-F1
3. **business/politics** sports F1 0.99 business politics 0.93~0.97 ” ”
4. “ ” ” ”
5. BoW 24 70 BERT 105 2

9.1

1.
2.
3. **BERT**

9.2 ” ”

Logistic Regression 110M BERT 49 Macro-F1 BERT 4 BERT ” ” ” ”

9.3

NYT 604 BERT max_length=64 49 8.0% ” ”

9.4

CPU BERT batch size “ ” ” ”

10.1

```
HW-1/
   homework1.py              # Task 1:
   homework1_word2vec.py     # Task 2.2 & 2.3: Word2Vec
   homework1_glove.py        # Task 2.1: GloVe
   homework1_bert.py         # Task 3: BERT
   nyt.csv                   # NYT
   ag.csv                    # AG News
   task1_results.csv         # Task 1
   task2_results.csv         # Task 2 Word2Vec
   task2_glove_results.csv   # Task 2 GloVe
   task3_results.csv         # Task 3
```

10.2

```
pip install numpy pandas scikit-learn gensim transformers torch accelerate
```

Task 1

```
python homework1.py
```

Task 2.1 GloVe glove.6B.100d.txt

```
#  https://nlp.stanford.edu/data/glove.6B.zip     glove.6B/
python homework1_glove.py
```

Task 2.2 & 2.3 Word2Vec

```
python homework1_word2vec.py
```

Task 3 BERT bert-base-uncased

```
#  https://huggingface.co/google-bert/bert-base-uncased     bert-base-uncased/
python homework1_bert.py
```

10.3

- 42
- BERT 440MB HuggingFace
- GloVe 400MB Stanford